

macro description

To enter a description about which macros are applied to an interface, use the **macro description** command in interface configuration mode. Use the **no** form of this command to remove the description. This command is mandatory for Auto SmartPorts to work.

macro description *text*
no macro description *text*

Syntax Description	description <i>text</i>	Enters a description about the macros that are applied to the specified interface.
Command Default	This command has no default setting.	
Command Modes	Interface configuration (config-if)	
Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	This command was introduced.
Usage Guidelines	Use the description keyword to associate comment text or the macro name with an interface. When multiple macros are applied on a single interface, the description text is from the last applied macro. You can verify your settings by entering the show parser macro description command in privileged EXEC mode.	

Example

This example shows how to add a description to an interface:

```
(config-if)# macro description duplex settings
```

clear ipv6 access-list

To reset the IPv6 access list match counters, use the **clear ipv6 access-list** command in privileged EXEC mode.

clear ipv6 access-list [*access-list-name*]

Syntax Description

<i>access-list-name</i>	(Optional) Name of the IPv6 access list for which to clear the match counters. Names cannot contain a space or quotation mark, or begin with a numeric.
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Command Default

No reset is initiated.

Command Modes

Privileged EXEC (#)

Command History

Release	Modification
Cisco IOS XE Everest 16.5.1a	This command was introduced.

Usage Guidelines

The **clear ipv6 access-list** command is similar to the **clear ip access-list counters** command, except that it is IPv6-specific.

The **clear ipv6 access-list** command used without the *access-list-name* argument resets the match counters for all IPv6 access lists configured on the router.

This command resets the IPv6 global ACL hardware counters.

Examples

The following example resets the match counters for the IPv6 access list named marketing:

```
# clear ipv6 access-list marketing
```

Related Commands

Command	Description
hardware statistics	Enables the collection of hardware statistics.
ipv6 access-list	Defines an IPv6 access list and enters IPv6 access list configuration mode.
show ipv6 access-list	Displays the contents of all current IPv6 access lists.

ip nat inside source

To enable Network Address Translation (NAT) of the inside source address, use the **ip nat inside source** command in global configuration mode. To remove the static translation, or the dynamic association to a pool, use the **no** form of this command.

Dynamic NAT

```
ip nat inside source { list { access-list-number access-list-name } | route-map name }
{ interface type number | pool name } [no-payload] [overload] [c] [vrf name]
no ip nat inside source { list { access-list-number access-list-name } | route-map name }
{ interface type number | pool name } [no-payload] [overload] [vrf name]
```

Static NAT

```
ip nat inside source static { interface type number | local-ip global-ip } [extendable] [no-alias]
[no-payload] [route-map name] [reversible] [vrf name] [forced]
no ip nat inside source static { interface type number | local-ip global-ip } [extendable]
[no-alias] [no-payload] [route-map name] [vrf name] [forced]
```

Port Static NAT

```
ip nat inside source static { tcp | udp } { local-ip local-port global-ip global-port } [extendable]
[forced] [no-alias] [no-payload] [route-map name] [vrf name] | interface global-port
no ip nat inside source static { tcp | udp } { local-ip local-port global-ip global-port } [extendable]
[forced] [no-alias] [no-payload] [route-map name] [vrf name] | interface global-port
```

Network Static NAT

```
ip nat inside source static network local-network global-network mask [extendable]
[forced] [no-alias] [no-payload] [vrf name]
no ip nat inside source static network local-network global-network mask [extendable] [forced]
[no-alias] [no-payload] [vrf name]
```

Syntax Description

list <i>access-list-number</i>	Specifies the number of a standard IP access list. Packets with source addresses that pass the access list are dynamically translated using global addresses from the named pool.
list <i>access-list-name</i>	Specifies the name of a standard IP access list. Packets with source addresses that pass the access list are dynamically translated using global addresses from the named pool.
route-map <i>name</i>	Specifies the named route map.
interface	Specifies an interface for the global address.
<i>type</i>	Interface type. For more information, use the question mark (?) online help function.
<i>number</i>	Interface or subinterface number. For more information about the numbering syntax for your networking device, use the question mark (?) online help function.
pool <i>name</i>	Specifies the name of the pool from which global IP addresses are allocated dynamically.

no-payload	(Optional) Prohibits the translation of an embedded address or port in the payload.
overload	(Optional) Enables the device to use one global address for many local addresses. When overloading is configured, the TCP or UDP port number of each inside host distinguishes between the multiple conversations using the same local IP address.
vrf <i>name</i>	(Optional) Associates the NAT translation rule with a particular VPN routing and forwarding (VRF) instance.
static	Sets up a single static translation.
<i>local-ip</i>	Local IP address assigned to a host on the inside network. The address could be randomly chosen, allocated from RFC 1918, or obsolete.
<i>global-ip</i>	Globally unique IP address of an inside host as it appears to the outside network.
extendable	(Optional) Extends the translation.
forced	(Optional) Forcefully deletes an entry and its children from the configuration.
no-alias	(Optional) Prohibits an alias from being created for the global address.
tcp	Establishes the TCP protocol.
udp	Establishes the UDP protocol.
<i>local-port</i>	Local TCP or UDP port. The range is from 1 to 65535.
<i>global-port</i>	Global TCP or UDP port. The range is from 1 to 65535.
network <i>local-network</i>	Specifies the local subnet translation.
<i>global-network</i>	Global subnet translation.
<i>mask</i>	IP network mask to be used with subnet translations.

Command Default No NAT translation of inside source addresses occurs.

Command Modes Global configuration (config)

Command History

Command History	Release	Modification
	Cisco IOS XE Gibraltar 16.10.1	This command was introduced.
	Cisco IOS XE Dublin 17.10.1	The route-map keyword was introduced.
	Cisco IOS XE Amsterdam 17.1.1	The vrf keyword was introduced.

Usage Guidelines The optional keywords of the **ip nat inside source** command can be entered in any order.

This command has two forms: the dynamic and the static address translation. The form with an access list establishes the dynamic translation. Packets from addresses that match the standard access list are translated using global addresses allocated from the pool named with the **ip nat pool** command.

Packets that enter the device through the inside interface and packets sourced from the device are checked against the access list for possible NAT candidates. The access list is used to specify which traffic is to be translated.

Alternatively, the syntax form with the keyword **static** establishes a single static translation.



Note When a session is initiated from outside with the source IP as the outside global address, the device is unable to determine the destination VRF of the packet.



Note When you configure NAT with a VRF-enabled interface address that acts as the global address, you must configure the **ip nat inside source static no-alias** command. If the **no-alias** keyword is not configured, Telnet to the VRF-enabled interface address fails.

Examples

The following example shows how to translate between inside hosts addressed from either the 192.0.2.0 or the 198.51.100.0 network to the globally unique 203.0.113.209/28 network:

```
ip nat pool net-209 203.0.113.209 203.0.113.222 prefix-length 28
ip nat inside source list 1 pool net-209
!
interface ethernet 0
 ip address 203.0.113.113 255.255.255.240
 ip nat outside
!
interface ethernet 1
 ip address 192.0.2.1 255.255.255.0
 ip nat inside
!
access-list 1 permit 192.0.2.1 255.255.255.0
access-list 1 permit 198.51.100.253 255.255.255.0
```

The following example shows how to translate the traffic that is local to the provider's edge device running NAT (NAT-PE):

```
ip nat inside source list 1 interface ethernet 0 vrf vrf1 overload
ip nat inside source list 1 interface ethernet 0 vrf vrf2 overload
!
ip route vrf vrf1 10.0.0.1 10.0.0.1 192.0.2.1
ip route vrf vrf2 10.0.0.1 10.0.0.1 192.0.2.1
!
access-list 1 permit 10.1.1.1 0.0.0.255
!
ip nat inside source list 1 interface ethernet 1 vrf vrf1 overload
ip nat inside source list 1 interface ethernet 1 vrf vrf2 overload
!
ip route vrf vrf1 10.0.0.1 10.0.0.1 198.51.100.1 global
ip route vrf vrf2 10.0.0.1 10.0.0.1 198.51.100.1 global
access-list 1 permit 10.1.1.0 0.0.0.255
```

The following example shows how to translate sessions from outside to inside networks:

```
ip nat pool POOL-A 10.1.10.1 10.1.10.126 255.255.255.128
ip nat pool POOL-B 10.1.20.1 10.1.20.126 255.255.255.128
ip nat inside source route-map MAP-A pool POOL-A reversible
```

```

ip nat inside source route-map MAP-B pool POOL-B reversible
!
ip access-list extended ACL-A
 permit ip any 10.1.10.128 0.0.0.127
ip access-list extended ACL-B
 permit ip any 10.1.20.128 0.0.0.127
!
route-map MAP-A permit 10
 match ip address ACL-A
!
route-map MAP-B permit 10
 match ip address ACL-B
!

```

The following example shows how to configure the route map R1 to allow outside-to-inside translation for static NAT:

```

ip nat inside source static 10.1.1.1 10.2.2.2 route-map R1 reversible
!
ip access-list extended ACL-A
 permit ip any 10.1.10.128 0.0.0.127
route-map R1 permit 10
 match ip address ACL-A

```

The following example shows how to configure NAT inside and outside traffic in the same VRF:

```

interface Loopback1
 ip vrf forwarding forwarding1
 ip address 192.0.2.11 255.255.255.0
 ip nat inside
 ip virtual-reassembly
!
interface Ethernet0/0
 ip vrf forwarding forwarding2
 ip address 192.0.2.22 255.255.255.0
 ip nat outside
 ip virtual-reassembly
ip nat pool MYPOOL 192.0.2.5 192.0.2.5 prefix-length 24
ip nat inside source list acl-nat pool MYPOOL vrf vrf1 overload
!
!
ip access-list extended acl-nat
 permit ip 192.0.2.0 0.0.0.255 any

```

Related Commands

Command	Description
access-list (IP extended)	Defines an extended IP access list.
access-list (IP standard)	Defines a standard IP access list.
clear ip nat translation	Clears dynamic NAT translations from the translation table.
interface	Configures an interface type and enters interface configuration mode.
ip access-list	Defines an IP access list or object group access control list by name or number.
ip nat	Designates that traffic originating from or destined for the interface is subject to NAT.

Command	Description
ip nat inside destination	Enables NAT of the inside destination address.
ip nat outside source	Enables NAT of the outside source address.
ip nat pool	Defines a pool of IP addresses for NAT.
ip nat service	Enables a port other than the default port.
ip route vrf	Establishes static routes for a VRF instance.
ip vrf forwarding	Associates a VRF instance with a diameter peer.
permit	Sets conditions in a named IP access list or object group access control list that will permit packets.
route-map	Defines the conditions for redistributing routes from one routing protocol into another routing protocol, or enables policy routing.
show ip nat statistics	Displays NAT statistics.
show ip nat translations	Displays active NAT translations.

ip igmp snooping vlan mrouter

To add a multicast router port, use the **ip igmp snooping mrouter** global configuration command on the stack or on a standalone . To return to the default settings, use the **no** form of this command.

Command Default	By default, there are no multicast router ports.
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Command Modes	Global configuration
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Command History	Release	Modification
	Cisco IOS XE Everest 16.5.1a	This command was introduced.

Usage Guidelines	VLAN IDs 1002—1005 are reserved for Token Ring and FDDI VLANs, and cannot be used in IGMP snooping. The configuration is saved in NVRAM.
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Example

The following example shows how to configure a port as a multicast router port:

```
(config)# ip igmp snooping vlan 1 mrouter interface gigabitethernet1/0/2
```

You can verify your settings by entering the **show ip igmp snooping** privileged EXEC command.

show license all

To display all licensing information enter the **show license all** command in privileged EXEC mode. This command displays status, authorization, UDI, and usage information, all combined.

show license all

Syntax Description

This command has no arguments or keywords.

Command Default

Privileged EXEC (#)

Command History

Release	Modification
Cisco IOS XE Fuji 16.9.1	This command was introduced.
Cisco IOS XE Amsterdam 17.3.2a	Command output was updated to display information relating to Smart Licensing Using Policy. Command output no longer displays Smart Account and Virtual account information.
Cisco IOS XE Cupertino 17.7.1	The output of the command was enhanced to display the following information: • RUM report statistics, in section <code>Usage Report Summary</code>. • Smart Account and Virtual Account information, in section <code>Account Information</code>.

Usage Guidelines

This command concatenates the output of other show license commands, enabling you to display different kinds of licensing information together. For field descriptions, refer to the corresponding commands.

Smart Licensing: If the software version on the device is Cisco IOS XE Amsterdam 17.3.1 or an earlier release, command output displays fields pertinent to Smart Licensing (whether smart licensing is enabled, all associated licensing certificates, compliance status, and so on).

Smart Licensing Using Policy: If the software version on the device (also referred to as a product instance) is Cisco IOS XE Amsterdam 17.3.2a or a later release, command output displays fields pertinent to Smart Licensing Using Policy.

- The `Smart Licensing Status` section corresponds with the output of the **show license status** command.
- The `License Usage` section corresponds with the output of the **show license usage** command.
- The `Product Information` section corresponds with the output of the **show license udi** command.
- The `Agent Version` section of the show license all command displays the Smart Agent version and is available only in this command.
- The `License Authorizations` section corresponds with the output of the **show license authorization** command.
- The `Usage Report Summary` section corresponds with the output in the **show license tech** command.